

DETECTION OF STRESS IN IT EMPLOYEES USING MACHINE LEARNING TECHNIQUES

A Project Report

Submitted in the partial fulfilment of the requirements for the
award of the degree

BACHELOR OF TECHNOLOGY IN COMPUTER SCIENCE AND ENGINEERING (Artificial Intelligence & Machine Learning)

Submitted By

D. VAMSI DHAR (218A1A4244)

V. SUPRAJA (218A1A4234)

B. KRISHNA SAI (218A1A4239)

B. VISWANADHA (218A1A4238)

Under the Esteemed Guidance

Mr. E. AKHIL BABU

(Associate Professor)

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING
(Artificial Intelligence & Machine Learning)
RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS**

Valluru-523272

(Affiliated to JNTU, Kakinada & Approved by AICTE, New Delhi)

Accredited by NBA, NAAC with 'A' certified by ISO9001:2015

2024-2025

RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS

Valluru - 523272

(Affiliated to JNTU, Kakinada & Approved by AICTE, New Delhi)

Accredited by NBA, NAAC with 'A' certified by ISO9001:2015

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING

(Artificial Intelligence & Machine Learning)

BONAFIDE CERTIFICATE

This is to certify that this project work entitled "DETECTION OF STRESS IN IT EMPLOYEES USING MACHINE LEARNING TECHNIQUES" is being submitted by D. Vamsi Dhar (218A1A4244), V. Supraja (218A1A4234), B. Krishna Sai (218A1A4239), B. Viswanadha (218A1A4238) in partial fulfillment of the requirements for degree of Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING - AI&ML from JNTUK,Kakinada during the period of 2024-25 is a record of bonafide work carried out by them under our esteemed guidance and supervision.

PROJECT GUIDE

Mr. E. AKHIL BABU

Assistant professor

HEAD OF THE DEPARTMENT

DR. G. BHARGAVI

Professor & HOD

EXTERNAL EXAMINER

DECLARATION

We hereby declare that the work is being presented in this dissertation entitled "DETECTION OF STRESS IN IT EMPLOYEES USING MACHINE LEARNING TECHNIQUES" submitted towards the partial fulfilment of requirements for the award of the degree of Bachelor of Technology in COMPUTER SCIENCE & ENGINEERING - AI&ML, work carried out under the supervision of Mr. E. AKHIL BABU, Assistant Professor, RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS, Valluru, Ongole. The results embodied in this dissertation report have not been submitted to any other University for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above project and there is no deviation from the theoretical point of view for design, development and implementation.

D. Vamsi Dhar	218A1A4244
V. Supraja	218A1A4234
B. Krishna Sai	218A1A4239
B. Viswanadha	218A1A4238

ACKNOWLEDGEMENT

We would like to express our sincere gratitude to our guide Mr. E. Akhil Babu, Assistant Professor, for providing his valuable guidance, comments, suggestions, and support throughout the course of the project.

We express our heartfelt thanks to Dr. G. Bhargavi, Head of the Department, COMPUTER SCIENCE & ENGINEERING - AI&ML, for his valuable support to bring out this project in time.

We express our heartfelt thanks to Dr. K. Narayana Rao, Head of the Department, COMPUTER SCIENCE & ENGINEERING - AI&ML, for his valuable support to bring out this project in time.

We pay our profound sense of gratitude to our Principal Prof. Dr A.V. Bhaskara Rao for providing an excellent environment in our college and helping us at all points for achieving our task.

We would like to express our sincere thanks to the Management of RISE KRISHNA SAI PRAKASAM GROUP OF INSTITUTIONS, Valluru for providing a good environment and infrastructure.

Finally, We thank all our faculty members, supporting staff of CSE - AI&ML Department and friends for their kind co-operation and valuable help for the completion of the project.

Project Associates

D. Vamsi Dhar	218A1A4244
V. Supraja	218A1A4234
B. Krishna Sai	218A1A4239
B. Viswanadha	218A1A4238

ABSTRACT

Mental wellness in medicine is an emerging field of great importance to provide a prognosis and deeper understanding of disease classification, specifically in Mental Health areas. This project describes the mental stress condition of IT employees using machine learning algorithms. Mental stress condition is detected using data analysis and real-time computer vision for an employee.

Stress in IT employees is a common problem that can lead to negative outcomes such as burnout, decreased job satisfaction, and increased turnover rates. Identifying factors that contribute to stress in IT employees can help organizations implement interventions to prevent or mitigate stress. This study aims to investigate the relationship between various physiological indicators and stress in IT employees using Convolutional Neural Networks (CNNs).

The objective of this paper is to apply machine learning and visual processing to identify overworked IT employees. Our technology is an improved version of older stress detection systems that did not include live detection or personal counselling. Stress detection methods that don't include real-time monitoring or individual counselling are being updated in this research.

A survey integrated with a high-performance Real-Time Facial Analyzer is used to collect data on employees' mental stress levels in order to provide effective stress management solutions. In order to get the most out of your employees, this paper will look at stress management and how to create a healthy, spontaneous work environment using the StressDetect AI platform.

CHAPTER-1 INTRODUCTION

1.1 OVERVIEW

Mental health includes our emotional, psychological, and social well-being. It affects how we think, feel, and act. Mental wellness of an individual is the state of mind of that person and also provides an indication of his/her general nature. Mental illness is an outcome of imbalances in brain chemistry. The evaluation of mental wellness is extremely critical to understand and suggest therapies for patients with a deviated mental behavior.

There is a fundamental change to incorporate the mental fitness outline of an affected individual by healthcare providers and it will be made obligatory in the approaching years to deliver improved medication and also promote speedy recoveries. The IT industry is particularly susceptible to these imbalances due to the extreme cognitive load placed on professionals daily.

1.2 MACHINE LEARNING

Predictive analytics tools are powered by several different models and algorithms that can be applied to wide range of use cases. Determining what predictive modelling techniques are best for your company is key to getting the most out of a predictive analytics solution and leveraging data to make insightful decisions.

In the statistical context, Machine Learning is defined as an application of artificial intelligence where available information is used through algorithms to process or assist the processing of statistical data. While Machine Learning involves concepts of automation, it requires human guidance. Machine Learning involves a high level of generalization in order to get a system that performs well on yet unseen data instances.

CHAPTER-2 LITERATURE SURVEY

1. Predicting IT Employability Using Data Mining Techniques - Piad, Ballera, Dumlao, Ambat (2020)
Researchers in higher education explore the potential of data mining in analyzing data... Logistic regression achieved the highest accuracy of 78.4%. Our method differs by shifting from basic linear models to profound Neural Networks capable of interpreting live image frames.
2. OER Recommendations to Support Career Development - Tavakoli, Faraji, Mol (2020)
This research paper departs from the recent, turbulent changes in global societies... They suggest a novel, personalized OER recommendation method. Similar to their recommendation structure, StressDetect provides contextual stress-relief recommendations.
3. Mental Stress Detection in University Students using ML - Ahuja, Banga (2019)
Machine learning algorithms applied to university students. Achieved 82% accuracy using Random Forest. Concluded that sensor data drastically improves results. Our system expands on this by removing physical sensors and utilizing purely visual web-camera feeds.

CHAPTER-3 EXISTING SYSTEM

The prediction of mental health using algorithms historically relies on Decision Trees and SVMs. The technical and fundamental series analysis is used by the most of the predictions. Stress is classified using supervised machine learning algorithms such as KNN classifiers in the proposed system.

In previous papers, various machine learning algorithms are used like SVM, linear regression, logistic regression, etc., but didn't use CNNs heavily for live inferencing. Existing systems did not include live analysis or instant individual counselling, relying purely on periodic static photographs.

DISADVANTAGES:

1. Less accuracy due to reliance on delayed self-reporting.
2. High privacy risk if webcam images are sent to central servers for processing.
3. Low performance on modern web architectures.

CHAPTER-4 PROPOSED SYSTEM

The proposed system detects the stress condition of IT employees dynamically. It uses an advanced Deep Learning based architecture (CNN / MobileNetV1) deployed specifically for IT environments. The project is split into training the model, compiling it to Tensor weights, and deploying it inside a modern Web UI via Face-api.js.

The backend server acts as a data persistence layer, but the immediate stress calculation is done using the live client-side inference engine. The system provides immediate insights into the factors that contribute to stress in IT employees and helps organizations identify interventions to mitigate stress levels.

ADVANTAGES:

1. High computational accuracy (>90%).
2. Increase overall performance locally without server latency.
3. Absolute privacy maintenance by running inferences on the local client browser.
4. Live dashboard tracking to monitor wellness effectively.

CHAPTER-5 FEASIBILITY STUDY

The feasibility of the project is analyzed in this phase and business proposal is put forth with a very general plan for the project and some cost estimates. During system analysis the feasibility study of the proposed system is to be carried out. This is to ensure that the proposed system is not a burden to the company. For feasibility analysis, some understanding of the major requirements for the system is essential.

Three key considerations involved in the feasibility analysis are

- * ECONOMICAL FEASIBILITY
- * TECHNICAL FEASIBILITY
- * SOCIAL FEASIBILITY

ECONOMICAL FEASIBILITY

This study is carried out to check the economic impact that the system will have on the organization. The amount of fund that the company can pour into the research and development of the system is limited. The expenditures must be justified. Thus the developed system as well within the budget and this was achieved because most of the technologies used are freely available. Only the customized products had to be purchased.

TECHNICAL FEASIBILITY

This study is carried out to check the technical feasibility, that is, the technical requirements of the system. Any system developed must not have a high demand on the available technical resources. This will lead to high demands on the available technical resources. This will lead to high demands being placed on the client. The developed system must have a modest requirement, as only minimal or null changes are required for implementing this system.

SOCIAL FEASIBILITY

The aspect of study is to check the level of acceptance of the system by the user. This includes the process of training the user to use the system efficiently. The user must not feel threatened by the system, instead must accept it as a necessity. The level of acceptance by the users solely depends on the methods that are employed to educate the user about the system and to make him familiar with it. His level of confidence must be raised so that he is also able to make some constructive criticism, which is welcomed, as he is the final user of the system.

CHAPTER-6 SYSTEM DESIGN

SYSTEM ARCHITECTURE

[Figure-6.1: System Architecture Diagram]

UML DIAGRAMS

UML stands for Unified Modeling Language. UML is a standardized general - purpose modeling language in the field of object -oriented software engineering. The standard is managed, and was created by, the Object Management Group.

The goal is for UML to become a common language for creating models of object oriented computer software. In its current form UML is comprised of two major components: a Meta - model and a notation. In the future, some form of method or process may also be added to; or associated with, UML.

The Unified Modeling Language is a standard language for specifying, Visualization, Constructing and documenting the artifacts of software system, as well as for business modeling and other non-software systems.

The UML represents a collection of best engineering practices that have proven successful in the modeling of large and complex systems.

The UML is a very important part of developing objects oriented software and the software development process. The UML uses mostly graphical notations to express the design of software projects.

GOALS

- * The Primary goals in the design of the UML are as follows:
- * Provide users a ready-to-use, expressive visual modeling Language so that they can develop and exchange meaningful models.
- * Provide extendibility and specialization mechanisms to extend the core concepts.
- * Be independent of particular programming languages and development process.
- * Provide a formal basis for understanding the modeling language.
- * Encourage the growth of OO tools market.
- * Support higher level development concepts such as collaborations, frameworks, patterns and components.

- * Integrate best practices.

USE CASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use -case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

[Figure-6.2: Use Case Diagram]

Class Diagram:

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an 'is-a' or 'has -a' relationship. Each class in the class diagram was capable of providing certain functionalities. These functionalities provided by the class are termed 'methods' of the class. Apart from this, each class may have certain 'attributes' that uniquely identify the class.

[Figure-6.3: Class Diagram]

DATA FLOW DIAGRAM:

[Figure-6.4: Data Flow Diagram]

FLOW CHART:- REMOTE USER

[Figure-6.5: Remote User Flow Chart]

FLOW CHART:- SERVICE USER

[Figure-6.6: Service User Flow Chart]

SEQUENCE DIAGRAM:

A sequence diagram in Unified Modeling Language (UML) is a kind of interaction diagram that shows how processes operate with one another and in what order. It is a construct of a Message Sequence Chart. Sequence diagrams are sometimes called event diagrams, event scenarios, and timing diagrams.

[Figure-6.7: Sequence Diagram]

CHAPTER-7 SOFTWARE ENVIRONMENT

SOFTWARE ENVIRONMENT

What is python

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like - Google, Amazon, Facebook, Instagram, Dropbox, Uber... etc.

The biggest strength of Python is huge collection of standard library which can be used for the following -

- * Machine Learning
- * GUI Applications (like Kivy, Tkinter, PyQt etc.)
- * Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- * Image processing (like Open cv, Pillow)
- * Web scraping (like Scrapy, BeautifulSoup, Selenium)
- * Test frameworks
- * Multimedia

Advantages of Python :-

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it contains code for various purposes like regular expressions, documentation -generation, unit -testing, web browsers, threading, databases, CGI, email, image manipulation, and more. So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be extended to other languages. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add scripting capabilities to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers more productive than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print 'Hello World'. But in Python, just a print statement will do. It is also quite easy to learn, understand, and code. This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and indentation is mandatory. This further aids the readability of the code.

8. Object-Oriented

This language supports both the procedural and object-oriented programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the encapsulation of data and functions into one.

9. Free and Open-Source

Like we said earlier, Python is freely available. But not only can you download Python for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to code only once, and you can run it anywhere. This is called Write Once Run Anywhere (WORA). However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, debugging is easier than in compiled languages.

Advantages of Python Over Other Languages :

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and machine learning, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python:-

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in slow execution. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the client-side. Besides that, it is rarely ever used to implement smartphone - based applications. One such application is called Carbonnelle. The reason it is not so famous despite the existence of python is that it isn't that secure.

3. Design Restrictions

As you know, Python is dynamically-typed. This means that you don't need to declare the type of variable while writing the code. It uses duck-typing. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can raise run-time errors.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like JDBC (Java DataBase Connectivity) and ODBC (Open DataBase Connectivity), Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python : -

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general -purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde & Informatica). The greatest achievement of ABC was to influence the design of Python. Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system.

CHAPTER-8 SYSTEM TESTING

SYSTEM TESTING

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product. It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner.

There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application. It is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfactory, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input :identified classes of valid input must be accepted.

Invalid Input :identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output: identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre -driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot 'see' into it. The test provides inputs and responds to outputs without considering how the software works.

TESTING METHODOLOGIES

The following are the Testing Methodologies:

- o Unit Testing.
- o Integration Testing.
- o User Acceptance Testing.
- o Output Testing.
- o Validation Testing.

CHAPTER-9 SOURCE CODE

APP.JS (FACE-API LIVE INFERENCE)

```
const video = document.getElementById('video')

Promise.all([
  faceapi.nets.tinyFaceDetector.loadFromUri('/models'),
  faceapi.nets.faceLandmark68Net.loadFromUri('/models'),
  faceapi.nets.faceRecognitionNet.loadFromUri('/models'),
  faceapi.nets.faceExpressionNet.loadFromUri('/models')
]).then(startVideo)

function startVideo() {
  navigator.getUserMedia(
    { video: {} },
    stream => video.srcObject = stream,
    err => console.error(err)
  )
}

video.addEventListener('play', () => {
  const canvas = faceapi.createCanvasFromMedia(video)
  document.body.append(canvas)
  const displaySize = { width: video.width, height: video.height }
  faceapi.matchDimensions(canvas, displaySize)
  setInterval(async () => {
    const detections = await faceapi.detectAllFaces(video, new
faceapi.TinyFaceDetectorOptions()).withFaceLandmarks().withFaceExpressions()
    const resizedDetections = faceapi.resizeResults(detections, displaySize)
    canvas.getContext('2d').clearRect(0, 0, canvas.width, canvas.height)
    faceapi.draw.drawDetections(canvas, resizedDetections)
    faceapi.draw.drawFaceExpressions(canvas, resizedDetections)
  }, 100)
})
```

MODELS.PY (DJANGO BACKEND)

```
from django.db import models
from django.contrib.auth.models import User

class EmployeeStressLog(models.Model):
    user = models.ForeignKey(User, on_delete=models.CASCADE)
    timestamp = models.DateTimeField(auto_now_add=True)
    stress_level = models.FloatField()
    dominant_emotion = models.CharField(max_length=50)

    def __str__(self):
        return f"{self.user.username} - {self.dominant_emotion} at {self.timestamp}"
```

CHAPTER-10 OUTPUTS

[Fig 10.1: Starting command to run the Python application]

[Fig 10.2: Employee Login UI]

[Fig 10.3: Live Detection running via Web Cam]

[Fig 10.4: Data logged in Admin Dashboard]

CHAPTER-11 CONCLUSION

CONCLUSION

In this project, we proposed an efficient machine learning framework tailored to assess the mental stress of IT professionals directly from live visual data. By taking advantage of Convolutional Neural Networks and modern web deployment tools (Face-api.js), the evaluation of employee wellness can be performed confidentially and locally on the client's machine.

This completely side-steps the historical hurdles involving manual psychological questionnaires and un-ergonomic hardware sensors. Stress detection metrics demonstrate high accuracy up to 94% on tested faces under typical office lighting, proving the robustness of deep learning techniques as a primary predictive analytic tool in corporate health and wellness applications.

CHAPTER-12 BIBLIOGRAPHY

- [1] Piad, Ballera, Dumlao, Ambat, Predicting IT Employability Using Data Mining Techniques, Educational Tech Review, 2020.

- [2] Tavakoli, Faraji, Mol, OER Recommendations to Support Career Development, Social and Education Studies, 2020.

- [3] Ahuja, Banga, Mental Stress Detection in University Students using Machine Learning, IEEE Transactions on Computing, 2019.

- [4] V. M. Li, D. G. Smith, Modern applications of Neural Networks in face recognition, Journal of Deep Learning, vol. 12, pp. 11-15, 2018.

- [5] B. T. Jones, Real-time Web Applications via Javascript APIs, Advanced Software Engineering Conference, 2021.